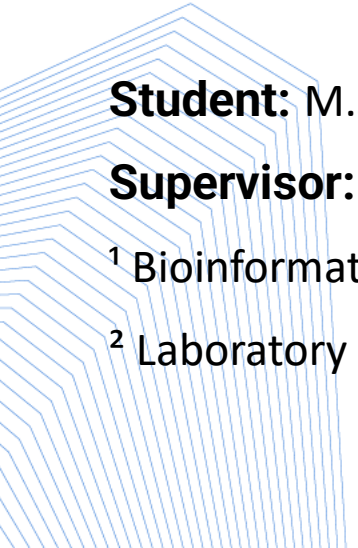


Reproducible sptPALM pipeline for CAZ protein mobility analysis



Student: M.S. Lantsova¹

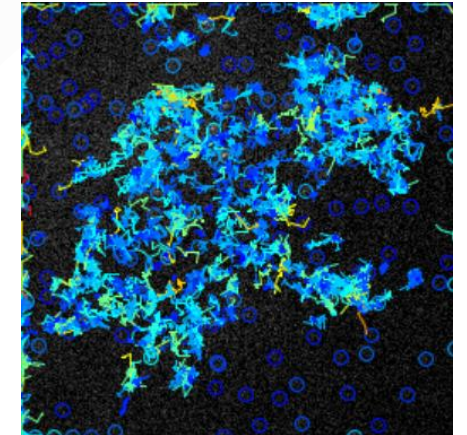
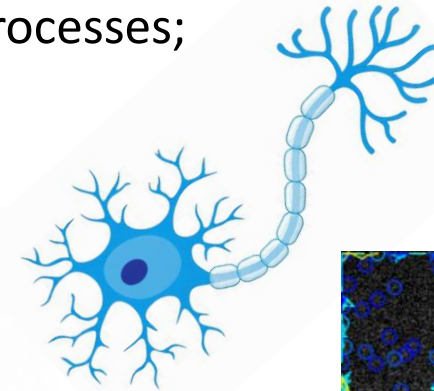
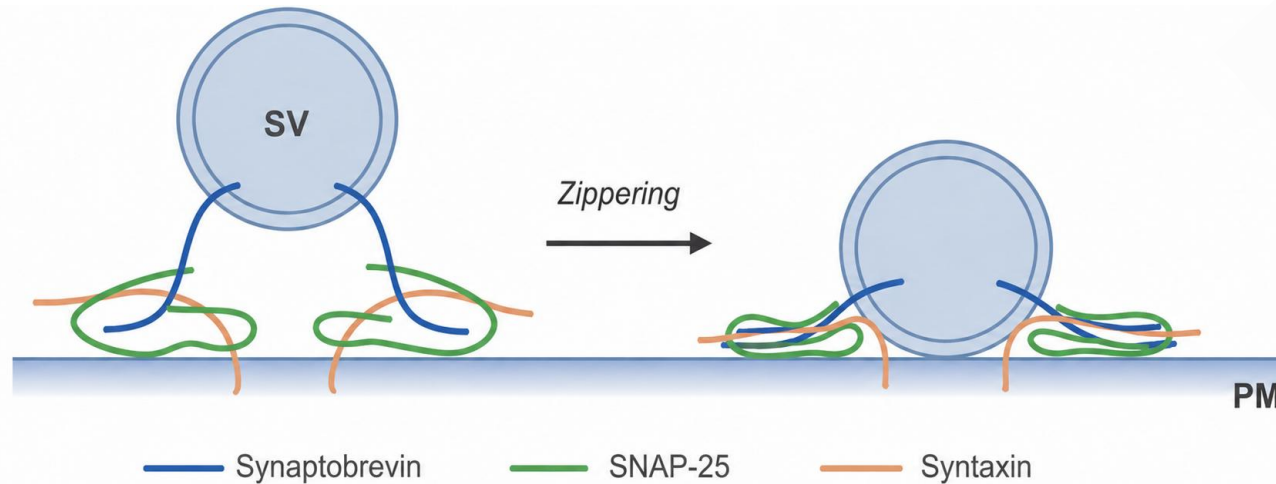
Supervisor: G.A. Nosov^{1,2}

¹ Bioinformatics Institute, Kantemirovskaya st. 2A, 197342, St. Petersburg, Russia

² Laboratory of Neurotechnology, Federal Center for Brain and Neurotechnologies, FMBA of Russia, Moscow, Russia

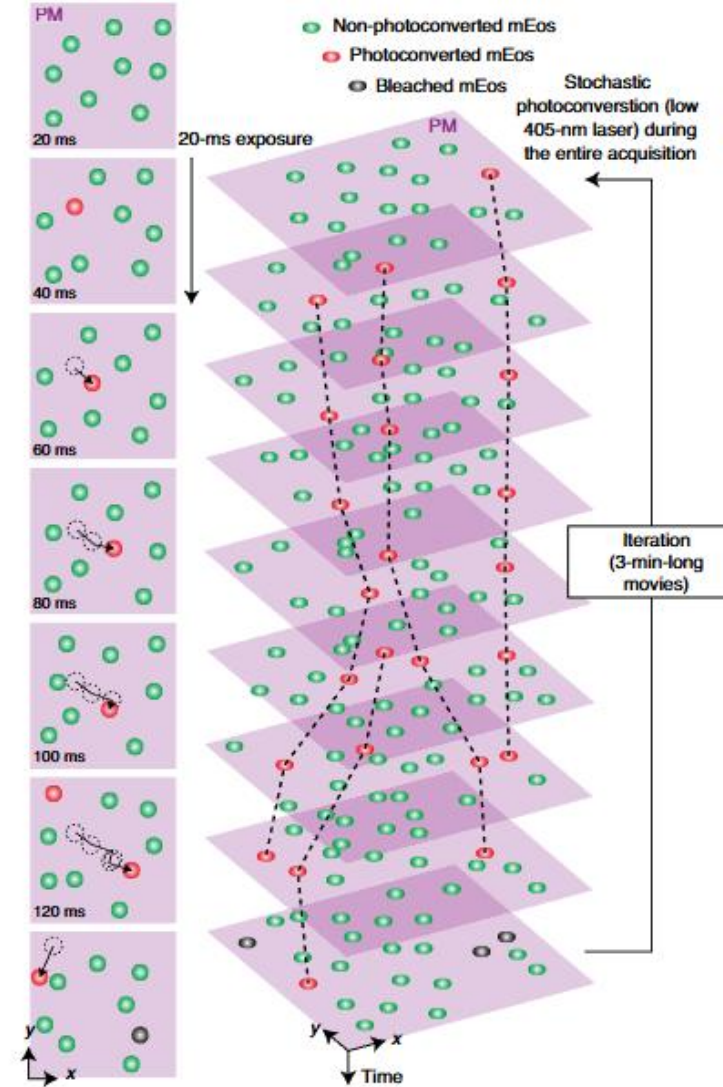
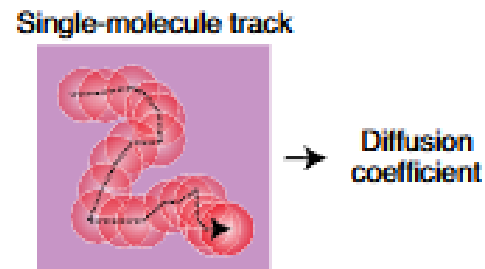
Synaptic protein mobility and membrane dynamics

- Synaptic proteins are dynamically redistributed within the plasma membrane;
- Oxidative stress may alter protein mobility and membrane organization;
- Single-molecule methods enable nanoscale analysis of these processes;



Single-particle tracking PALM (sptPALM)

- ❖ Sparse photoactivation;
- ❖ Single-molecule localization;
- ❖ Trajectory reconstruction;
- ❖ Quantitative mobility analysis;



Pipeline: Localization → Tracking → MSD → Diffusion → States

Single-particle tracking PALM (sptPALM)

Main goal:

Develop a reproducible computational pipeline for quantitative sptPALM analysis of synaptic protein dynamics under oxidative stress.

Objectives:

- ✓ preprocess TrackMate trajectories;
- ✓ remove tracking artifacts;
- ✓ estimate diffusion coefficients;
- ✓ analyze velocity distributions;
- ✓ classify mobility states;
- ✓ implement HMM inference;
- ✓ evaluate pipeline robustness;

Experimental design

Proteins:

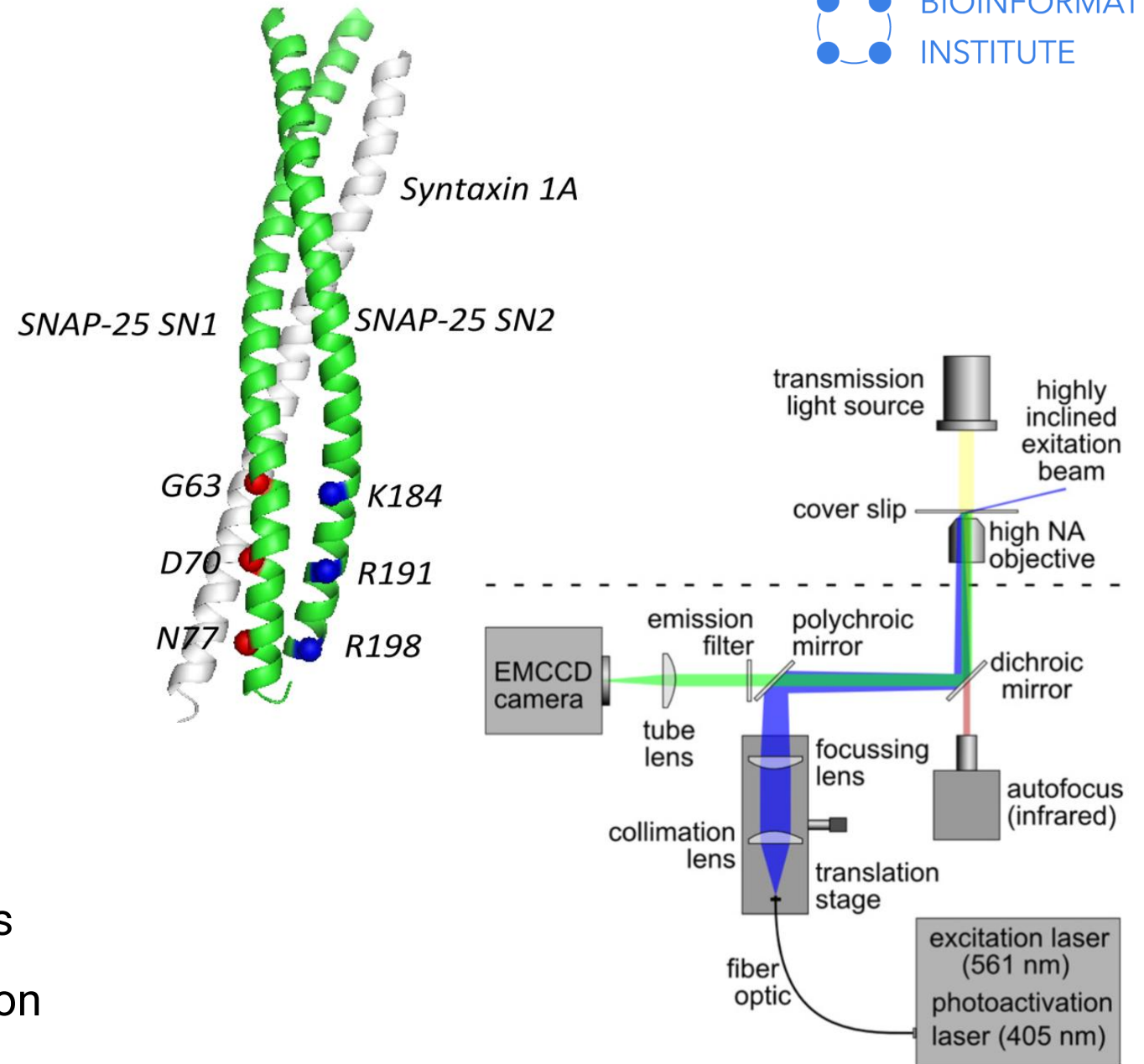
- SNAP25b
- Syntaxin

Conditions:

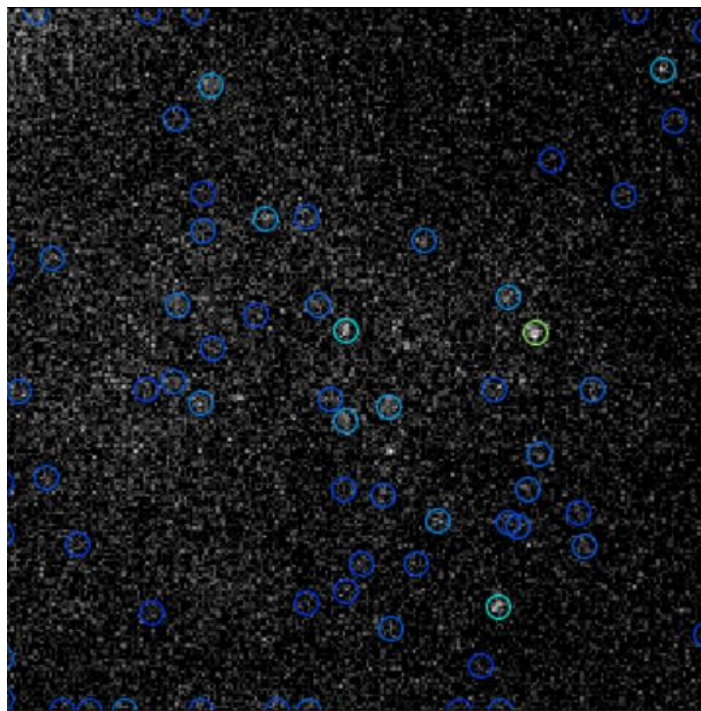
- Control
- 500 μM H_2O_2
- 500 μM H_2O_2 30 min

Imaging:

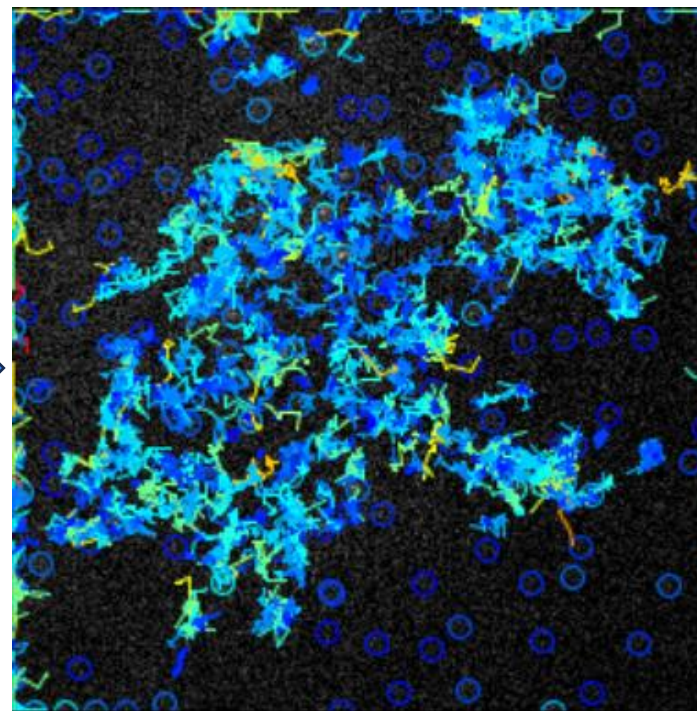
- TIRF microscopy
- 405 / 488 / 561 nm lasers
- single-molecule acquisition



Trajectory acquisition

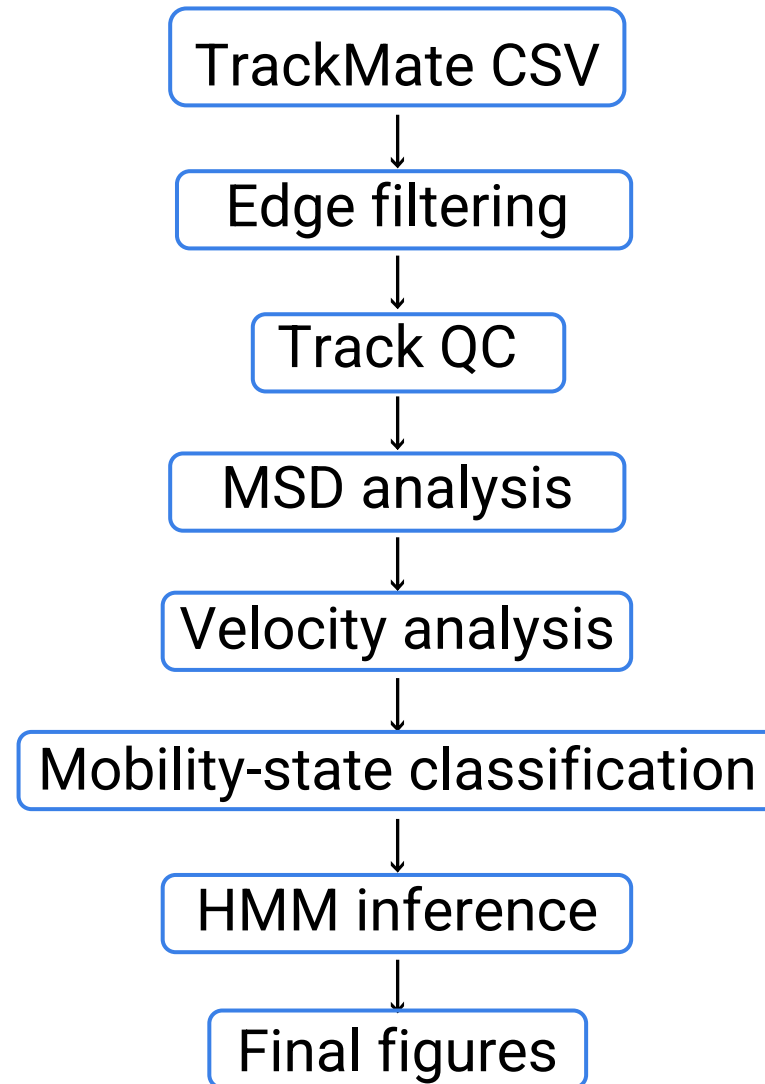


localization points



reconstructed tracks

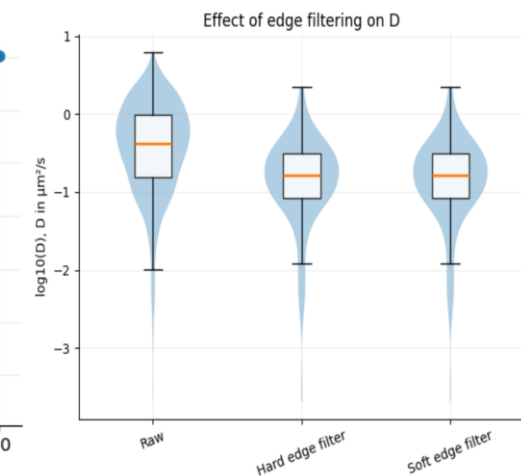
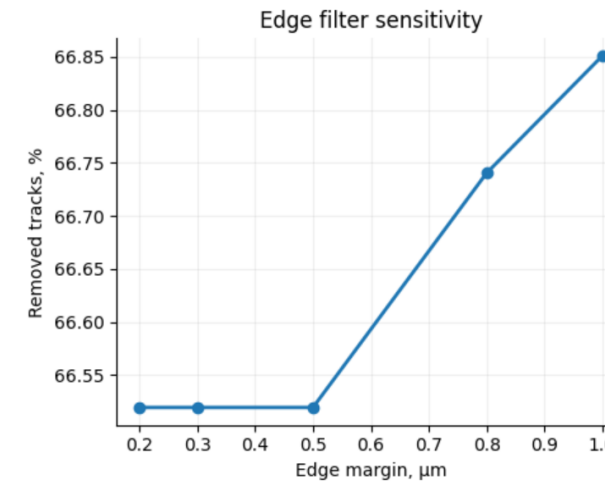
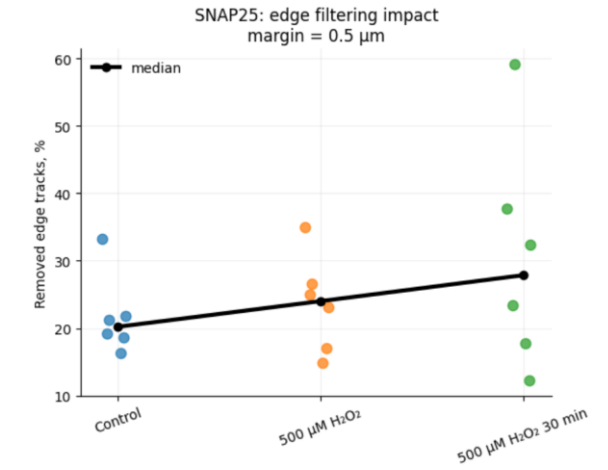
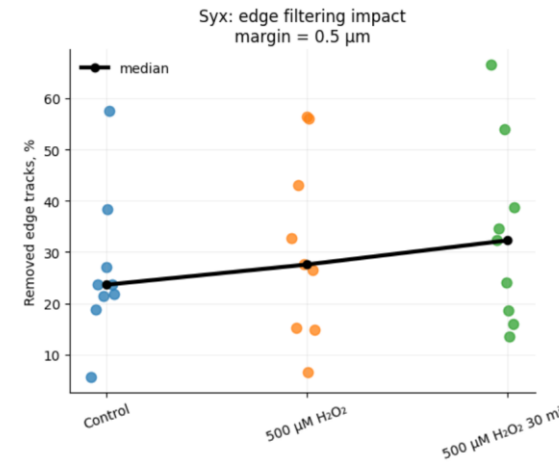
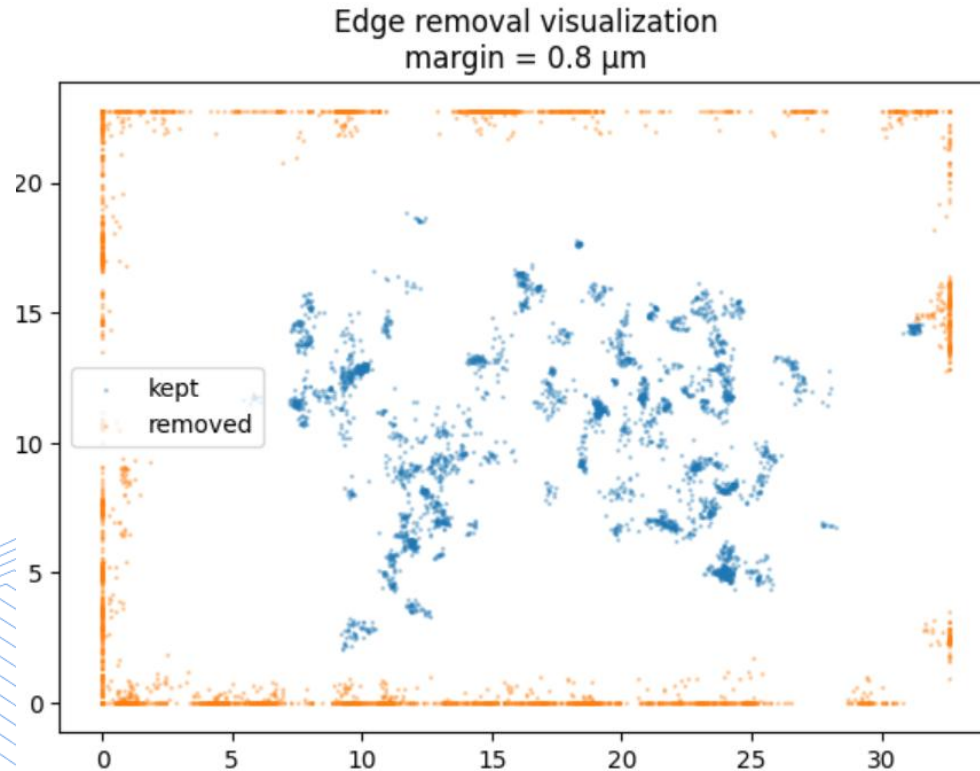
Processing pipeline



Python + pandas + numpy + TrackMate + hmmlearn

Edge-associated tracking artifacts

Large fractions of non-biological trajectories accumulated near image borders

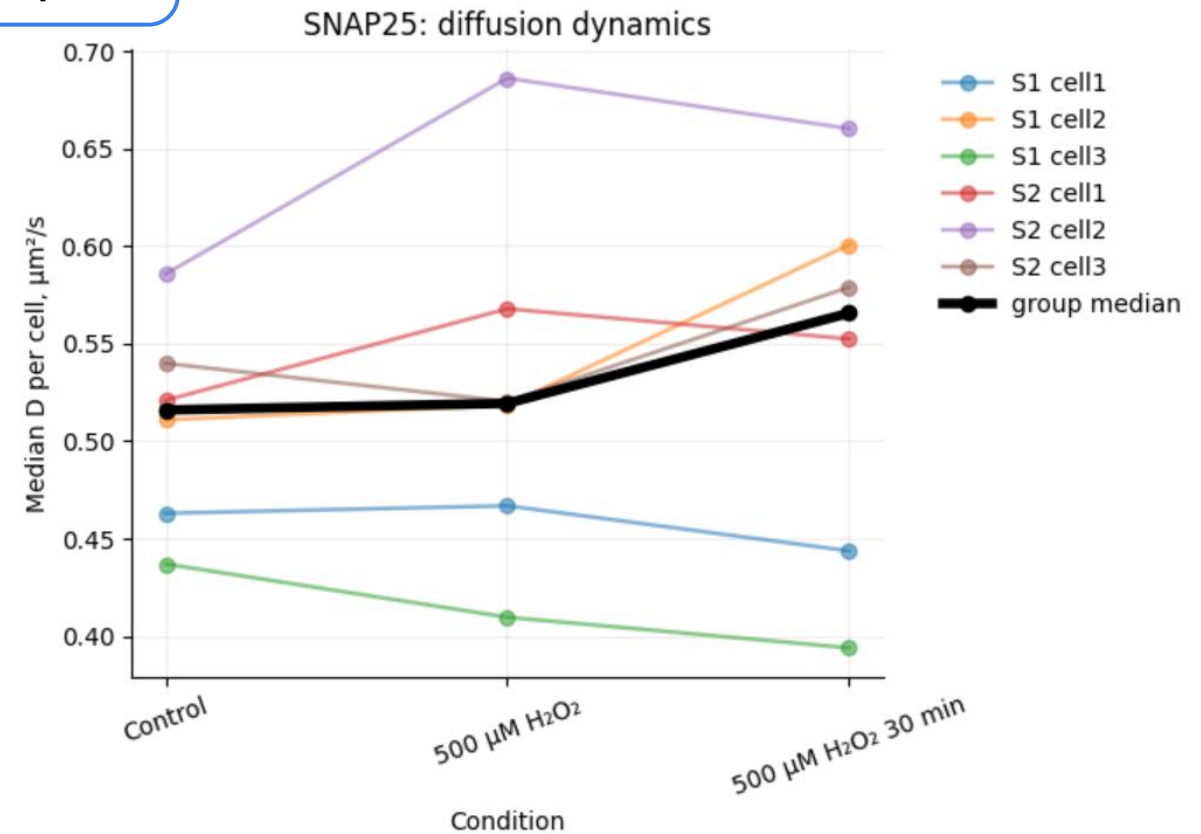
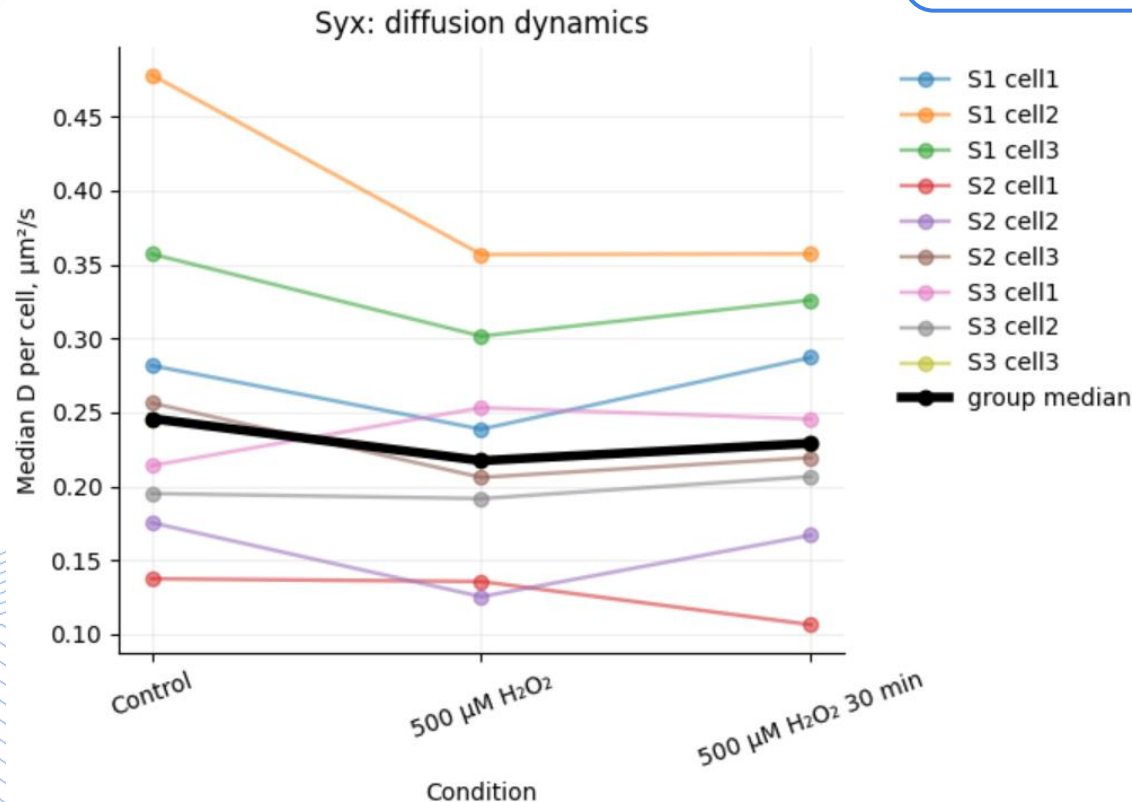


Edge filtering significantly reduced artifact influence on diffusion estimates

Diffusion dynamics

$$\text{MSD}(\tau) = 4D\tau$$

$$D = \text{slope} / 4$$

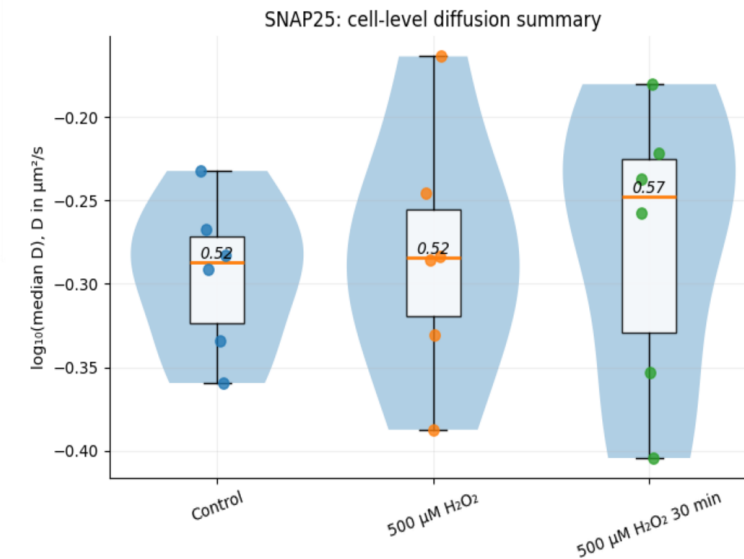
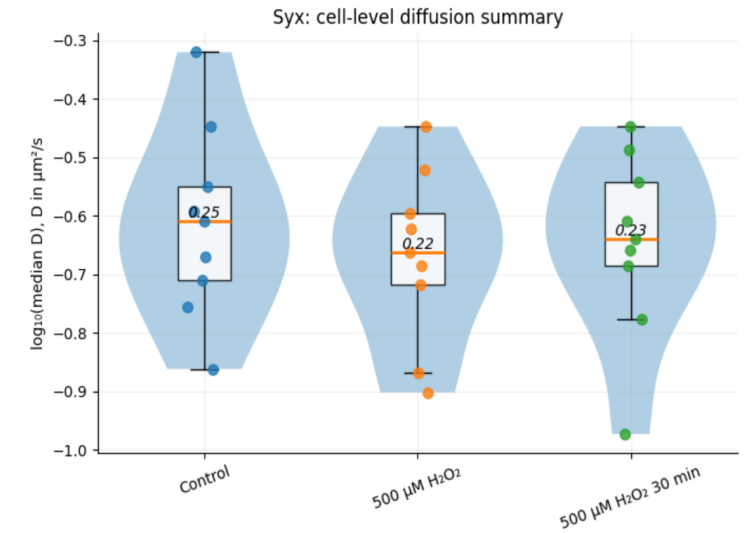
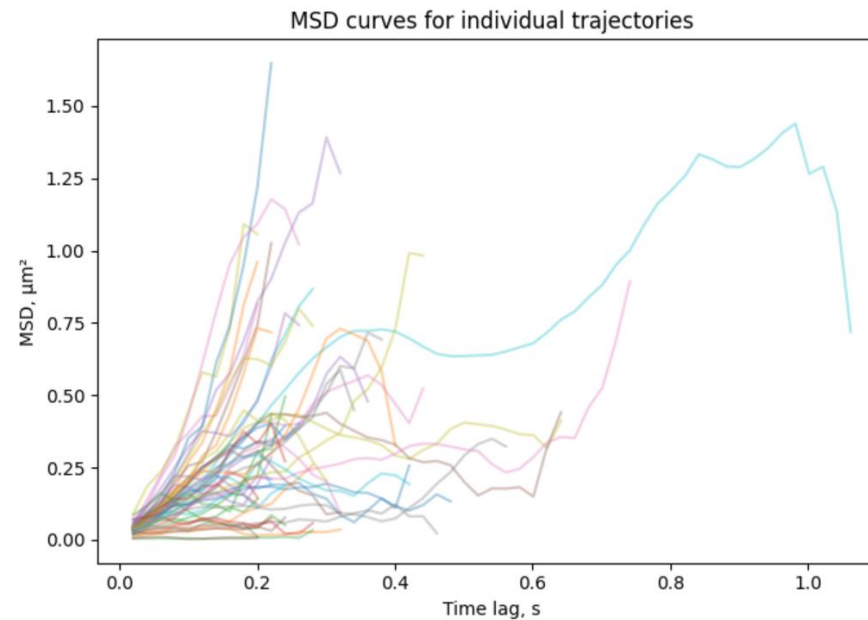
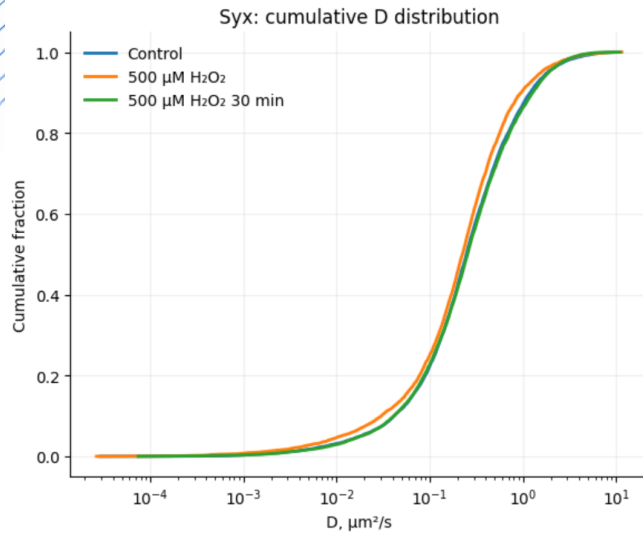


- heterogeneous diffusion behavior observed
- oxidative stress altered diffusion distributions

Diffusion dynamics

$$\text{MSD}(\tau) = 4D\tau$$

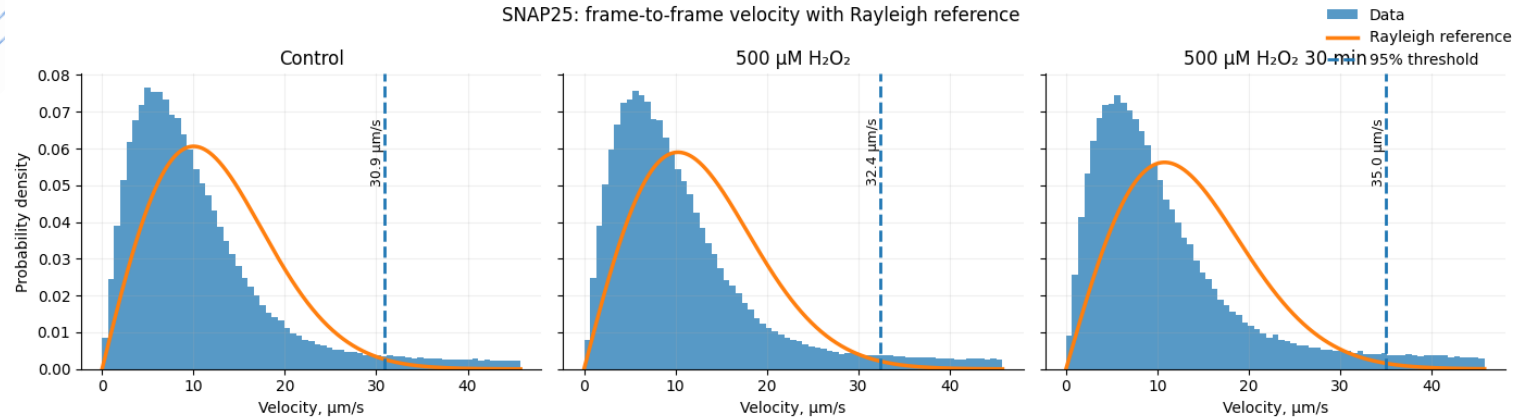
$$D = \text{slope} / 4$$



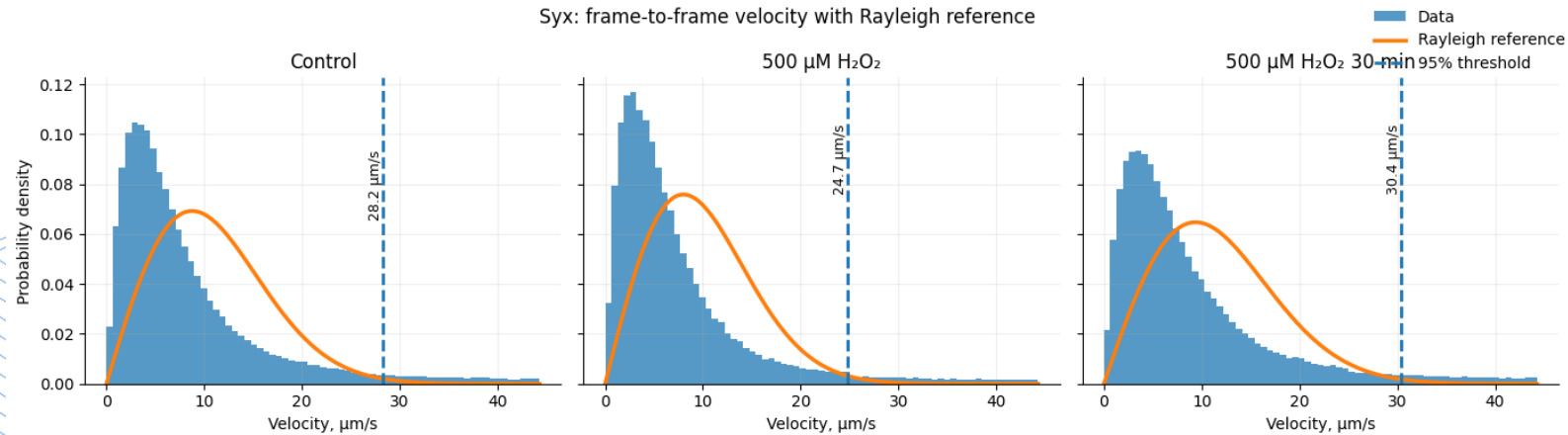
- heterogeneous diffusion behavior observed
- oxidative stress altered diffusion distributions

Frame-to-frame velocity analysis

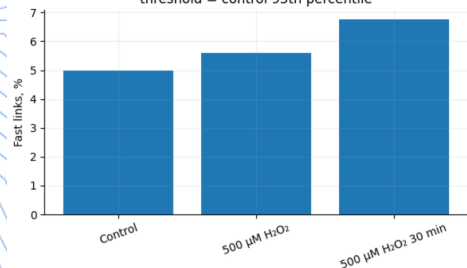
SNAP25: frame-to-frame velocity with Rayleigh reference



Syx: frame-to-frame velocity with Rayleigh reference

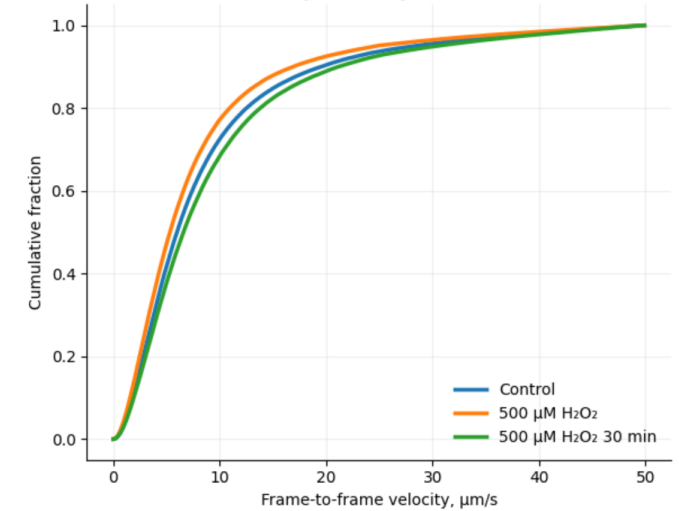


SNAP25: high-velocity tail threshold = control 95th percentile

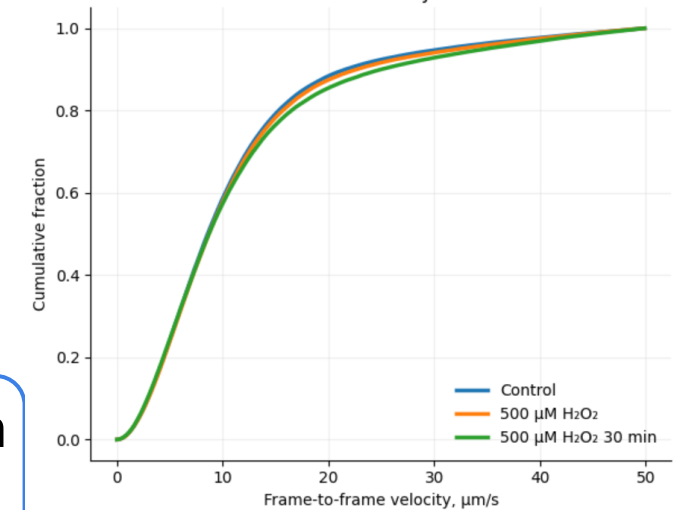


- distributions deviate from ideal Brownian diffusion
- heavy tails indicate heterogeneous mobility states

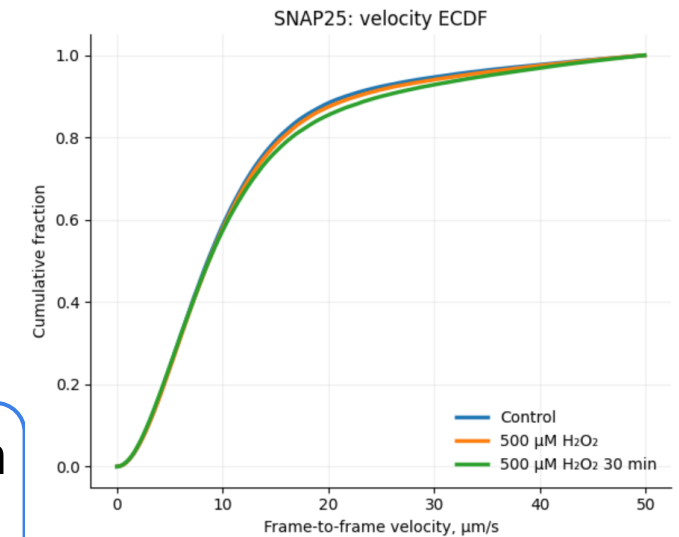
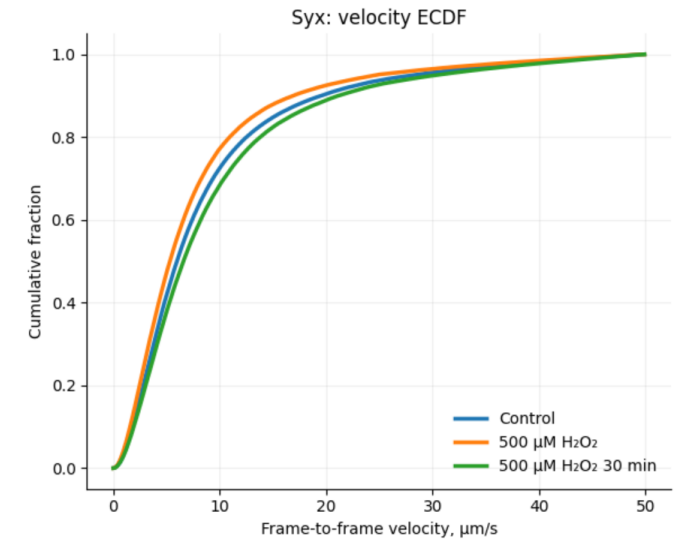
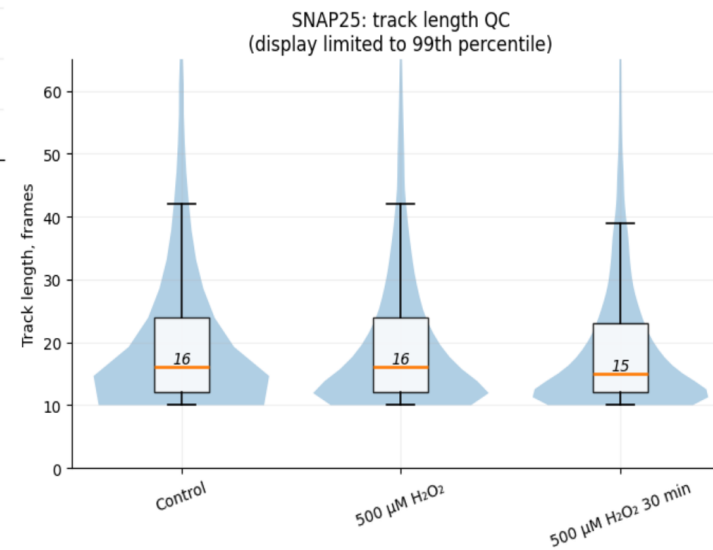
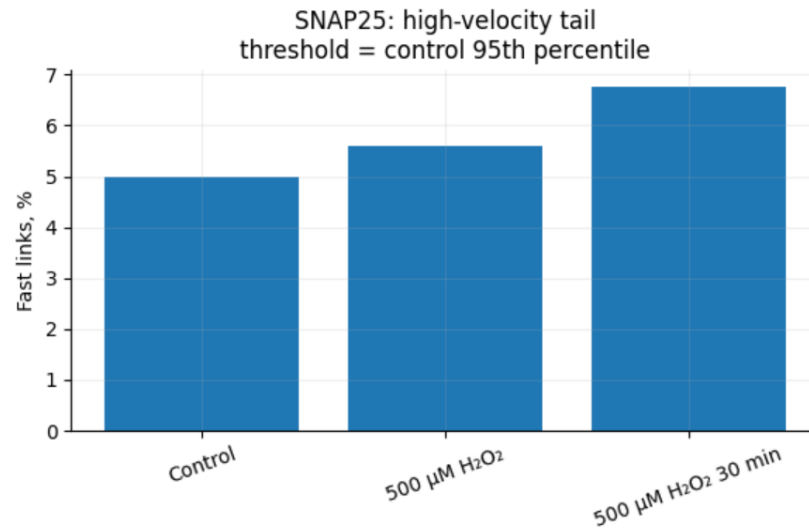
Syx: velocity ECDF



SNAP25: velocity ECDF

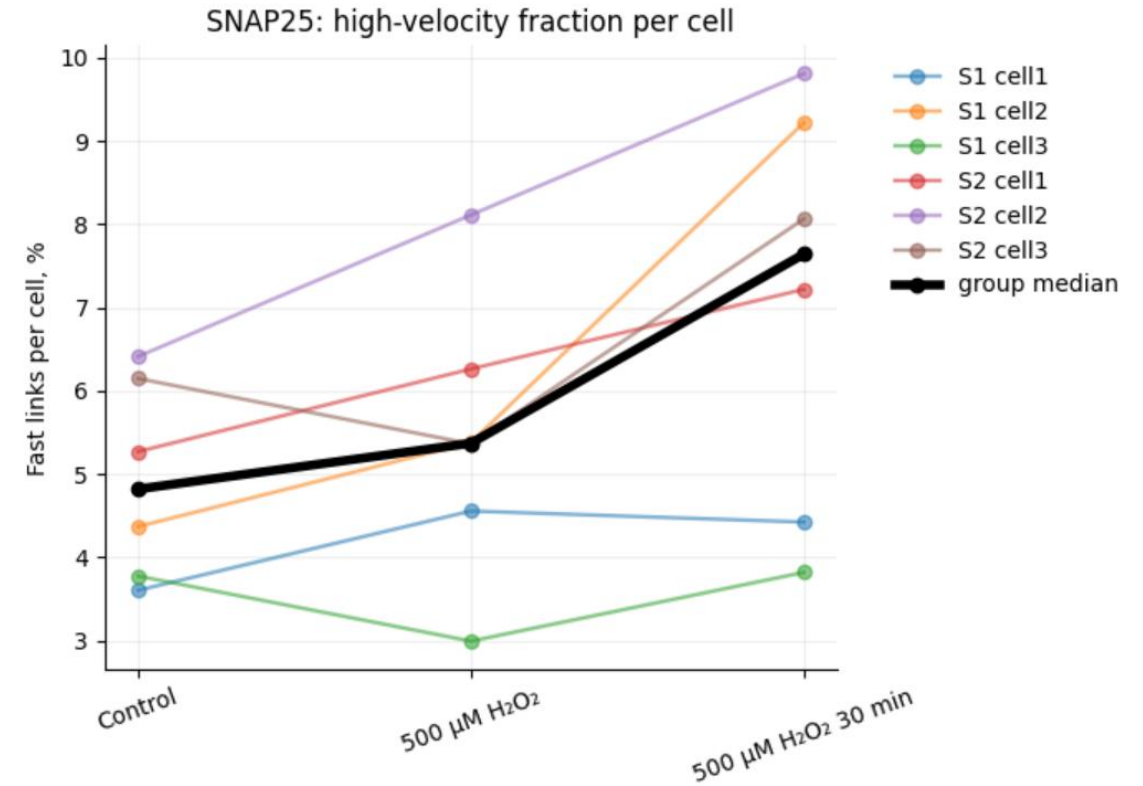
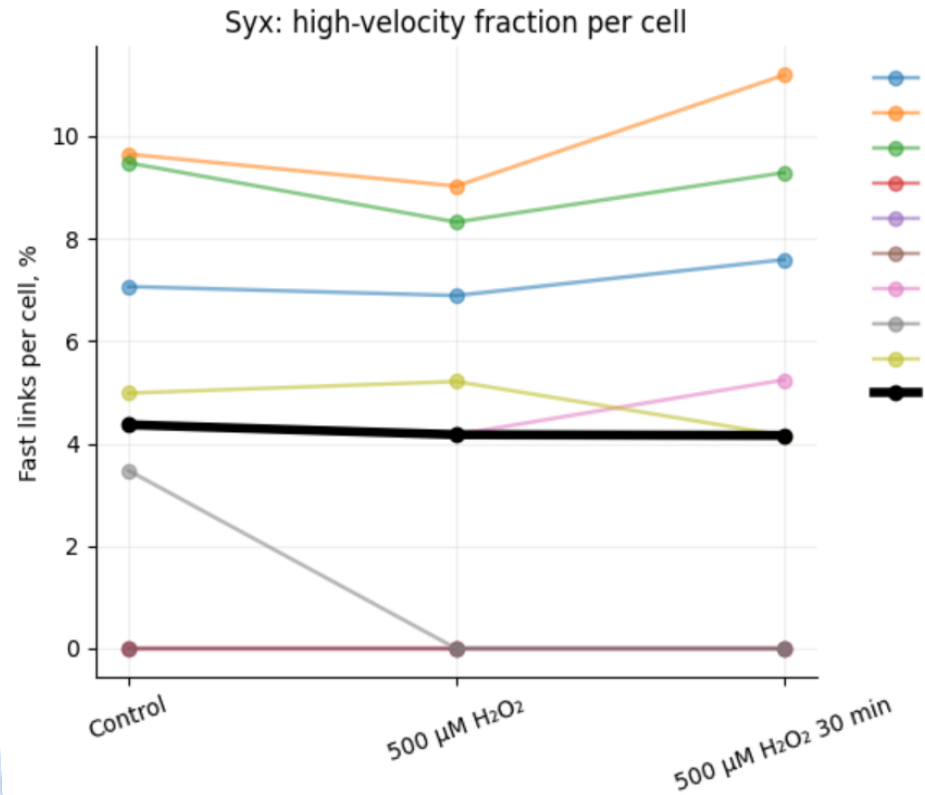


Frame-to-frame velocity analysis



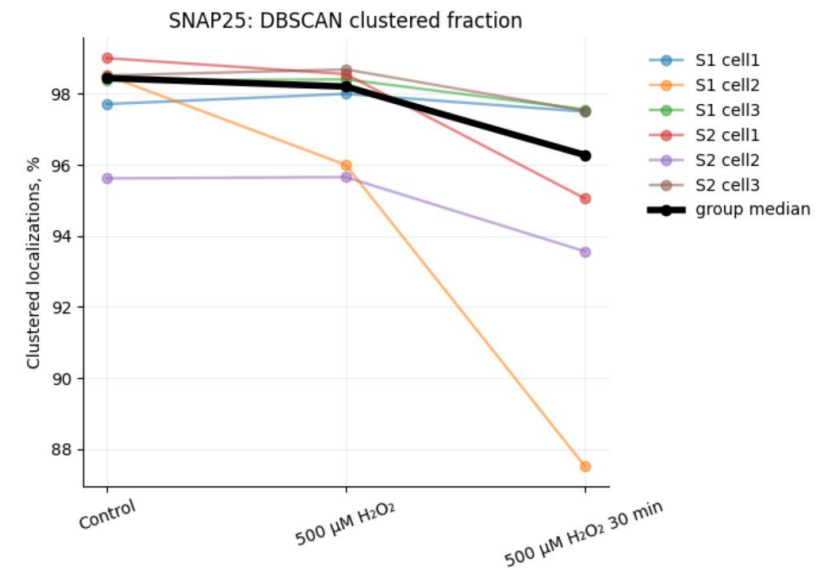
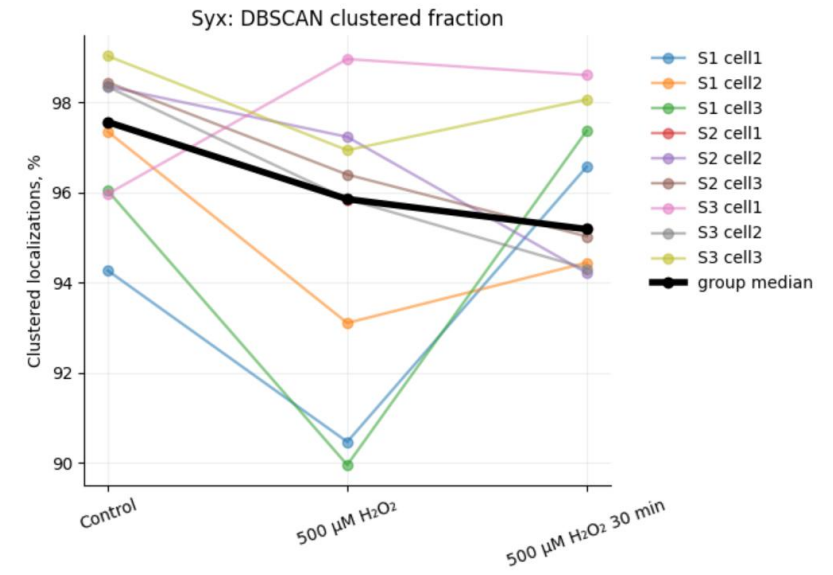
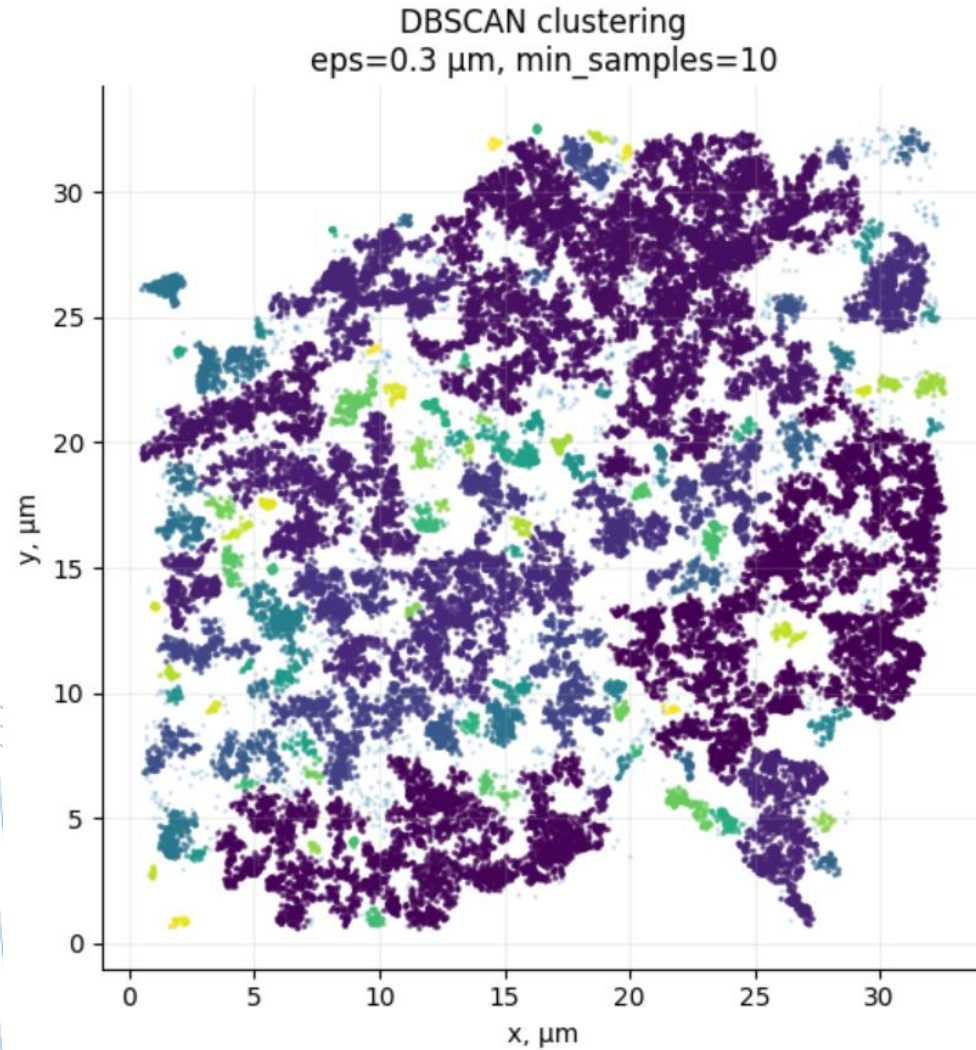
- distributions deviate from ideal Brownian diffusion
- heavy tails indicate heterogeneous mobility states

Mobility redistribution under oxidative stress



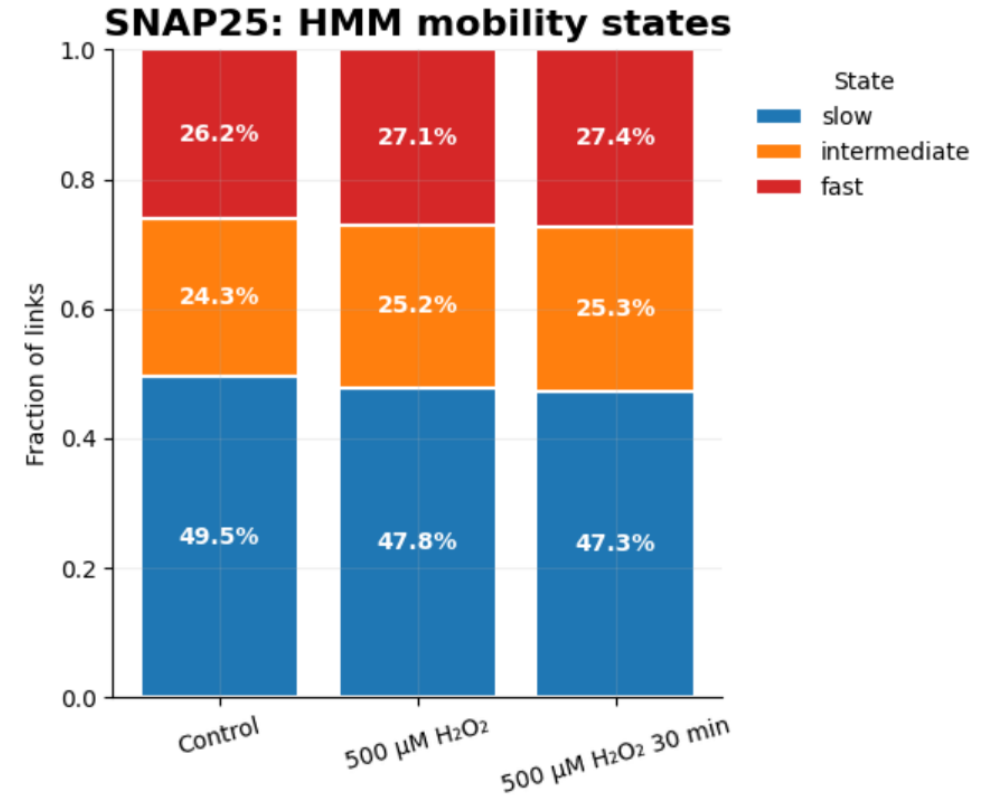
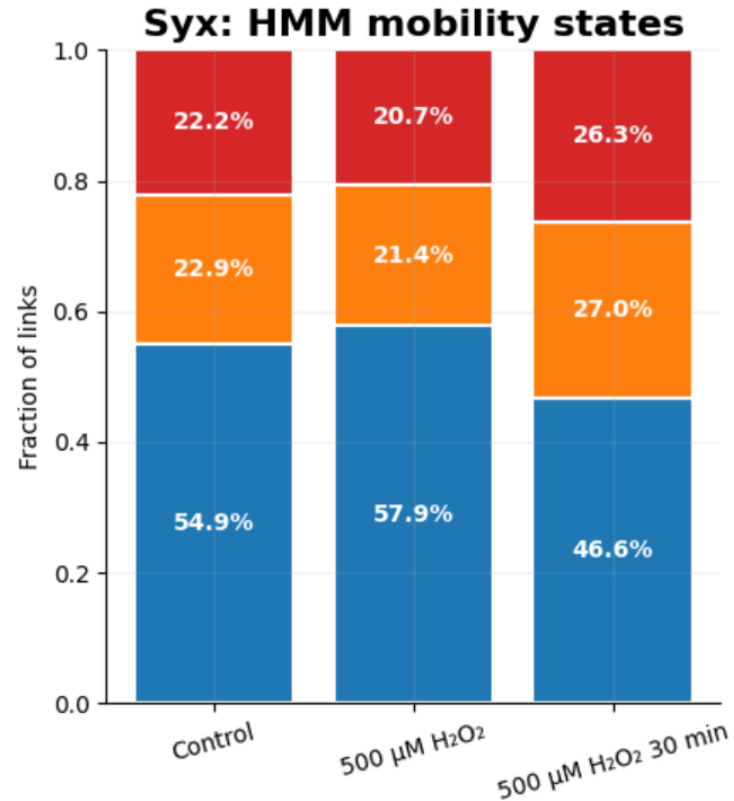
- prolonged H₂O₂ exposure increased highly mobile SNAP25 fraction
- protein-specific response observed

Spatial clustering analysis with DBSCAN



Hidden Markov Model inference

Slow ↔ Intermediate ↔ Fast



- preliminary HMM analysis revealed multiple dynamic states
- transitions between mobility states can be inferred computationally

Open and reproducible analysis

Repository structure:

sptPALM-analysis-of-CAZ-proteins/

- README.md
- requirements.txt
- data/
- notebooks/
- src/
- figures/
- results/
- docs/

Repository includes:

- preprocessing notebooks
- analysis modules
- visualization scripts
- reproducible figure generation

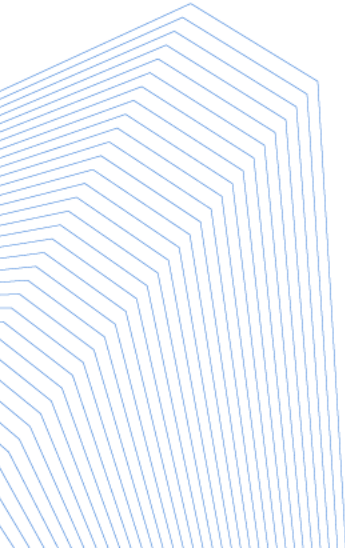


<https://github.com/MargaritaLantsova/sptPALM-analysis-of-CAZ-proteins>

Conclusions

- A reproducible computational workflow for sptPALM analysis was developed;
- Oxidative stress altered mobility distributions of synaptic proteins;
- Strong edge-associated artifacts were identified and corrected;
- Velocity distributions deviated from ideal Brownian behavior;
- HMM and clustering approaches demonstrated feasibility for future extensions;

Future directions:

- fully automated tracking
 - advanced HMM inference
 - spatial clustering analysis
 - additional synaptic proteins
 - neuronal datasets
- 
- 

Thank you!



Margarita Lantsova

MargaritaLantsova · she/her

Neurobiology @RNRMU @lift @msu

Lantsovams@gmail.com

tg: @Lantsovams21



Single-particle tracking PALM analysis of CAZ proteins in neuronal synapses

Student: Margarita Lantsova **Supervisor:** Georgii Nosov

Goal: Develop a reproducible computational pipeline for single-particle tracking PALM analysis of CAZ protein mobility and dynamics under different cellular conditions.

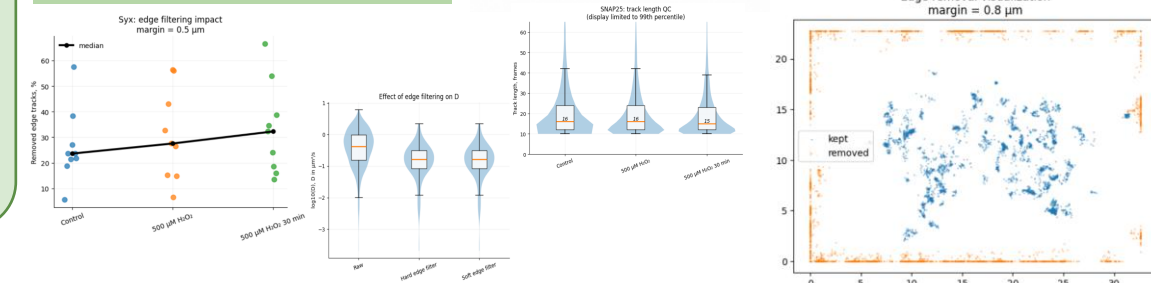
Data acquisition

- HEK293T / neurons
- CAZ protein expression (SNAP25b и Syntaxin)
- sptPALM imaging
- TrackMate export

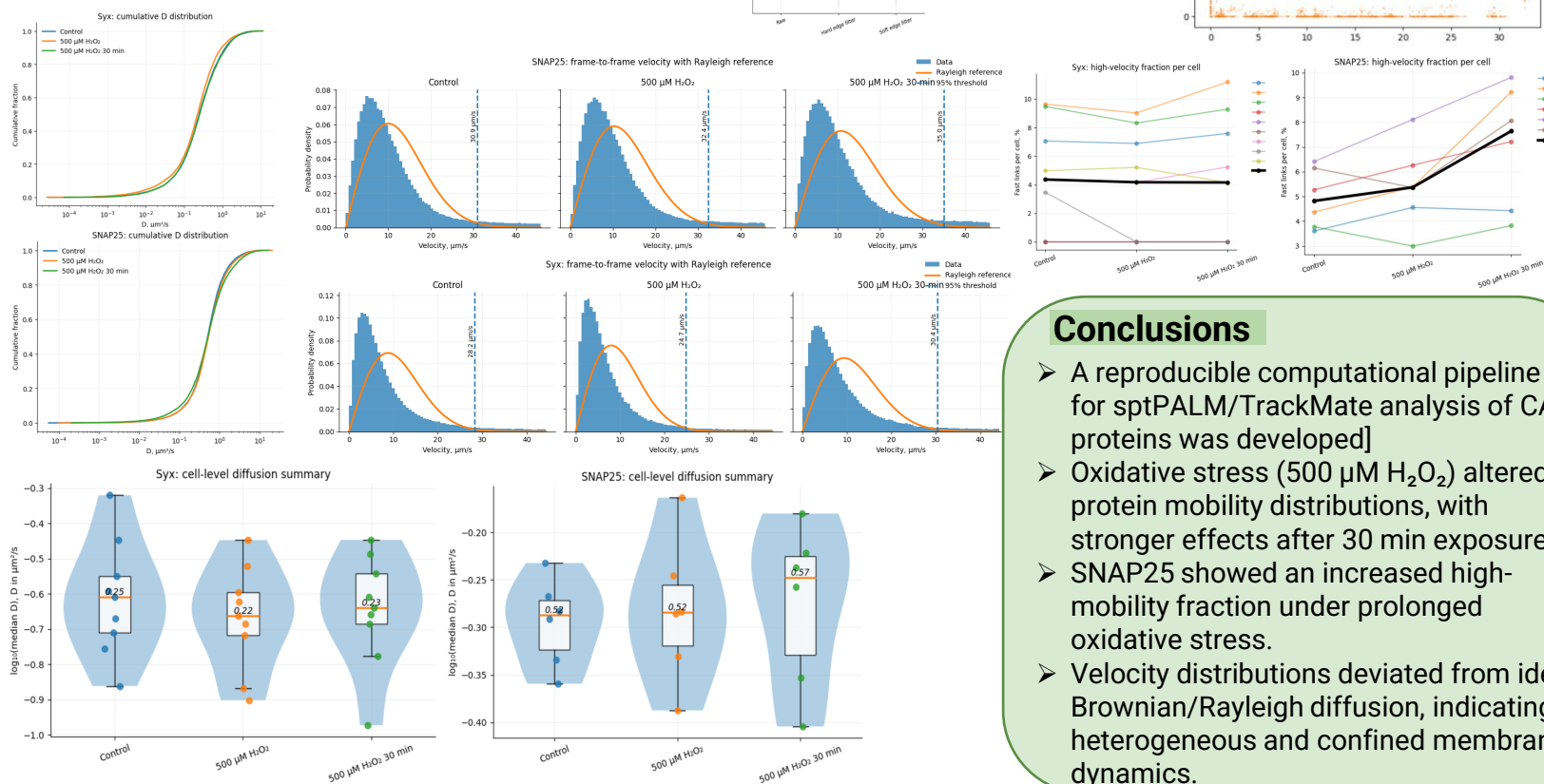
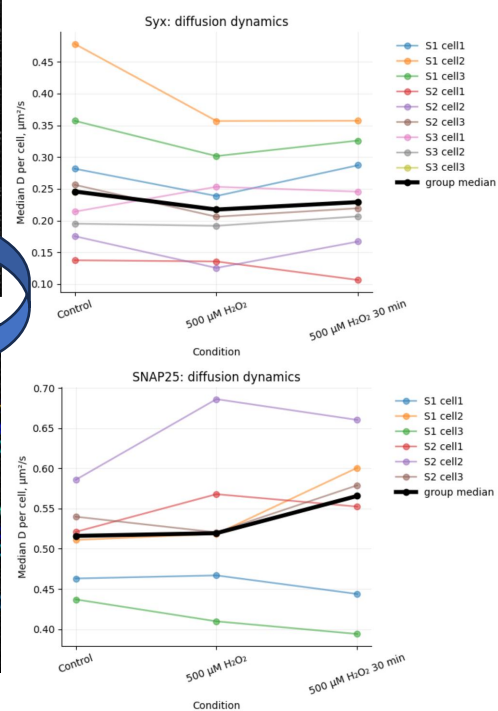
Processing pipeline

- **Particle detection & trajectory linking** (TrackMate CSV)
- **Edge filtering** (remove border artifacts)
- **Track QC** (length / displacement control)
- **MSD / D estimation** ($D = \text{slope} / 4$)
- **State classification** (slow / mobile fractions)
- **Statistics & visualization** (paired comparison across conditions)

QC and artifact control



Results



Conclusions

- A reproducible computational pipeline for sptPALM/TrackMate analysis of CAZ proteins was developed]
- Oxidative stress (500 μM H_2O_2) altered protein mobility distributions, with stronger effects after 30 min exposure.
- SNAP25 showed an increased high-mobility fraction under prolonged oxidative stress.
- Velocity distributions deviated from ideal Brownian/Rayleigh diffusion, indicating heterogeneous and confined membrane dynamics.